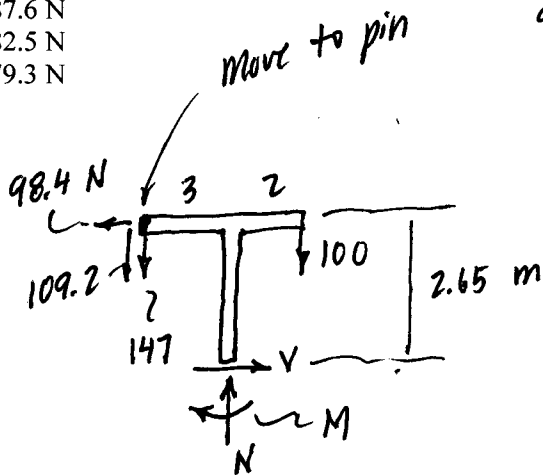


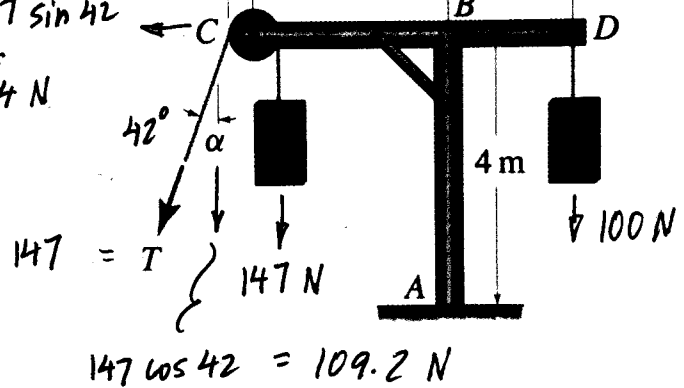
Exam 3: Practice Exam

1. (4 pt) Determine the magnitude of the shear force at a horizontal section on frame ABCD that lies 1.35 m above fixed support A. Load W is 147 N, $\alpha = 42^\circ$ and the counterweight Q is 100 N.

- A. 98.4 N
B. 92.1 N
C. 87.6 N
D. 82.5 N
E. 79.3 N



$$147 \sin 42^\circ = 98.4 \text{ N}$$



$$147 \cos 42^\circ = 109.2 \text{ N}$$

By inspection: $V = 98.4 \text{ N}$

2. (4 pt) Determine the magnitude of the normal force at a horizontal section on frame ABCD that lies 1.35 m above fixed support A. Load W is 147 N, $\alpha = 42^\circ$ and the counterweight Q is 100 N.

- A. 356 N
B. 362 N
C. 368 N
D. 374 N
E. 383 N

By inspection:

$$N = 100 + 147 + 109.2$$

$$N = 356.2 \text{ N} \quad (C)$$

3. (5 pt) Determine the magnitude of the bending moment at a horizontal section on frame ABCD that lies 1.35 m above fixed support A. Load W is 147 N, $\alpha = 42^\circ$ and the counterweight Q is 100 N.

- A. 829 N-m
B. 838 N-m
C. 846 N-m
D. 812 N-m
E. 801 N-m

$$+\circlearrowleft \sum M_{cut} = 0; \quad M + 100(2) - 98.4(2.65) - (147 + 109.2)(3) = 0$$

$$M = 829.4 \text{ N}\cdot\text{m}$$

